

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

7th September 2021

Subject CS1A – Actuarial Statistics (Paper A)

Time allowed: 3 Hours 30 Minutes (9.30 - 13.00 Hours)

Total Marks: 100

- Q. 1)** i) Define three “key forms of data analysis” using examples. (6)
- ii) Which of the following is not a property for a data set to be classified as “big data”?
- A. Speed of arrival of data
 - B. Size of the data set
 - C. Homogeneity within data set
 - D. Reliability of data elements may be difficult to ascertain
- (1)
[7]
- Q. 2)** The number of hospitalisations due to COVID19 have varied experience over months and data corresponding to a group illustrates that one should expect:
- Two hospitalisations in October
Three hospitalisations in November
One hospitalisation in December
- Determine the probability that there will be less than five hospitalisations in the period October to December given the hospitalisation in each month are independent and are assumed to follow Poisson distribution. [5]
- Q. 3)** The amount of claims of a motor insurance company is modelled as an exponential random variable with $\lambda = 1.25$ (in ‘000s). A data analyst is interested in assessing the probability of Y exceeding 10 (INR 10,000 if represented in absolute amounts), wherein; Y is the total of 10 independent motor claim amounts.
- i) Show, using moment generating functions, that:
- a) Y has gamma distribution and (3)
 - b) $2.5Y$ has a χ^2_{20} distribution (3)
- ii) Estimate the probability of $Y > 10$:
- A. 0.7986
 - B. 0.2014
 - C. 0.2140
 - D. None of the above
- (2)
- iii) Specify an approximate normal distribution for Y by applying the central limit theorem and use this to calculate the approximate value of $Y > 10$. (4)
- iv) Comment on the use of this approximation and on the result. (1)
[13]
- Q. 4)** A teacher wanted to analyse if there has been any alteration in the marks of his nine students after taking coaching classes. A test is conducted before students started the coaching classes and after completing the coaching classes. The scores before and after the coaching classes are given below:

Student #	1	2	3	4	5	6	7	8	9
Before coaching classes	50	35	37	49	40	38	44	40	43
After coaching classes	47	34	33	54	42	17	42	31	44

Determine the Pearson's correlation coefficient between the scores before and after the coaching classes:

- A. 0.8870
- B. 0.6795
- C. 0.7417
- D. 0.5421
- E. None of the above

[4]

Q. 5) Let X and Y be random variables with joint density function as

$$f(x, y) = (1/27) * (2x + y) \text{ and } X \text{ and } Y \text{ can assume values } 0, 1, 2$$

- i) Calculate the marginal distribution for Y. (3)
- ii) Compute the conditional distribution of $X=2$ given $Y=1$:

- A. 0.4
- B. 0.3333
- C. 0.5556
- D. 0.6667
- E. None of the above

(2)

[5]

- Q. 6)** i) State the properties of a “good estimator” and “consistent estimator”. (2)
- ii) In a particular small organisation, there are 5 Senior grade employees out of total 11 employees. You are told that ESOPs are granted to 3 employees.

Fisher's test – Summary of data provided.

ESOP	Yes	No	Total
Senior Grade	A	C	5
Other	B	D	6
Total	3	8	11

Four possible ways to fill the grid above are represented by say, $W1$, $W2$, $W3$ and $W4$ having probabilities of 0.0606, 0.1212, 0.3637 and 0.4545, respectively.

$W1$ to $W4$ are defined below: e.g., for $W1$, $A=3$, $B=0$, $C=2$ and $D=6$

<u>W 1</u>	<u>W 2</u>	<u>W 3</u>	<u>W 4</u>
3 2	0 5	2 3	1 4
0 6	3 3	1 5	2 4

- a) Using the information provided above, calculate the p-value of finding two senior grade employees having ESOPs. (2)

- b) Show that probability of $W1$ i.e., probability of finding 3 senior grade employees having ESOPs is 0.0606. (2)
- iii) The number of claims per year arising from Group-A of policies has a Poisson distribution with mean ' μ '. The number of claims arising from Group-B of policies has a Poisson distribution with mean 3 times ' μ '.

A sample of 100 Group-A policies resulted in 12 claims in a year and a sample of 80 Group-B policies resulted in 18 claims in a year. Determine the maximum likelihood estimate (MLE) of ' μ ' based on this information by selecting the option

- A. 0.088
 B. 0.120
 C. 0.167
 D. 0.225
 E. 0.367 (3)

- iv) A student has calculated MLE of the parameters of a lognormal distribution. Derive MLE of the mean and variance of the lognormal distribution if ' $\mu \text{ cap}$ '=1 and ' $\sigma^2 \text{ cap}$ ' = 0.5. (3)

- v) You are told that MLE of λ is 0.1 based on sample of 20 values with total delay of 200 (resulting in $\bar{X} = 10$).

MLE of 0.1 is obtained by setting first derivative of log likelihood with respect to λ equal to zero.

Second derivative of log likelihood with respect to λ is negative and was found to be $-n/\lambda^2$.

- a) Calculate large-sample approximate variance (i.e. CRLB) of ' $\lambda \text{ cap}$ '. (2)
- b) Further, calculate an approximate 95% confidence interval for ' λ '. (2)
- c) Calculate an exact 95% confidence interval for ' λ ' using chi square result and comment on the result in comparison to part (b) above. (4)

You are told that

$2*\lambda*n*\bar{X} \sim \text{Chi square distribution with } 2 \text{ times } n \text{ degrees of freedom}$
 24.43 and 59.34 are chi square 2.5% critical points

[20]

- Q. 7) i) The annual number of claims arising from a group of policies follows a Poisson distribution with mean ' μ '. The Prior distribution of ' μ ' is Ga (α , λ). In previous 5 years, the number of claims arising from the group were 3. Determine the posterior distribution of ' μ ':
- A. Ga ($\alpha+3$, $\lambda+5$)
 B. Ga ($\alpha+5$, $\lambda+3$)
 C. Ga (α , $\lambda+3$)

- D. $Ga(\alpha, \lambda + 5)$
 E. None of the above (2)

ii) If the prior distribution of 'mu' is Chi square distribution with 10 degrees of freedom and likelihood is Poisson with 'x' claims in 12 months, then identify the posterior distribution of 'mu'. (1)

iii) You are required to estimate 'mu' under all-or-nothing loss using posterior distribution as identified in part i.

Identify the correct option based on following statements-

- Your friend (F1) thinks that 'mu' would have been higher if observations would have been taken over longer time horizon irrespective of number of observed claims.
- Friend (F2) thinks that 'mu' would have been higher if more claims would have been observed over the same time horizon.

- A. F1 is definitely correct
 B. F2 is definitely correct
 C. Both F1 and F2 are definitely correct
 D. Both F1 and F2 are definitely wrong (1)

iv) Posterior distribution of theta is Beta (5,15) distribution. Identify the correct option indicating Bayesian estimate under all-or-nothing loss function:

- A. 5/15
 B. 5/20
 C. 4/19
 D. 4/18
 E. None of the above (3)

v) Total claim amount per annum on a particular insurance policy follows a normal distribution with unknown mean 'theta' and variance 150^2 . Prior beliefs about theta are described by normal distribution with mean 500 and variance 100^2 . Claim amounts $x_1, x_2, \dots, x_n$ are observed over n years.

a) State A, B, C and D for the posterior distribution of 'theta' which can be written in the form of $\text{Normal}((A+B)/(C+D), 1/(C+D))$ using values provided above. (2)

b) Express mean of the posterior distribution of 'theta' in terms of credibility estimate by clearly defining Z, prior mean and MLE of 'theta'. (2)

c) Comment on change in value of Z, using Z as derived in part b above:
 • if prior variance was 150^2
 • if likelihood variance was 100^2 (4)

[15]

Q. 8) You are an actuarial student working for a large life insurance company in India. Appointed Actuary has asked you to analyse the claims data to understand the impact of age on number of COVID claims. You have received the following information from claims department.

Age(X)	Number of COVID claims per 10,000 policies(Y)
5	101
15	120
25	135
35	186
45	268
55	540
65	620

$$\sum (x_i - \bar{x})^2 = 2,800 \quad \sum (y_i - \bar{y})^2 = 2,70,832 \quad \sum ((x_i - \bar{x})(y_i - \bar{y})) = 25,300$$

You have been asked to perform linear regression analysis on the data to identify the relationship between age and number of COVID claims.

- i) Determine the fitted regression line with 'no. of claims' as the response and 'age' as the explanatory variable. (3)
- ii) Assuming the full normal model, calculate the estimate of the error variance σ^2 and obtain a 90% confidence interval for σ^2 . (3)
- iii) Calculate the proportion of variance explained by the model. Hence, comment on the fit of the model. (3)
- iv) By considering the slope parameter, formally test whether the data is positively correlated. (3)
- v) Calculate 95% confidence interval for mean predicted number of COVID claims corresponding to age of 60. (4)
- vi) Assess the fitness of the model by:

- a) Completing the table of residuals (nearest rounded number)

Age	5	15	25	35	45	55	65
Residual	91		-56	-95		78	

(2)

- b) Comment on appropriateness of linear model. (3)

[21]

- Q. 9)** i) State three components of Generalised Linear Model (GLM). (2)

- ii) Show that exponential distribution with density function $f(y) = \frac{1}{\mu} e^{-\frac{y}{\mu}}$, $y > 0$ can be written in the form of exponential family of distribution. (1)

- iii) Further, determine.

- a) canonical link function (1)

- b) variance function (1)

- c) dispersion parameter (1)

Medical claim amounts of a health insurance company are believed to have an exponential distribution with mean μ_i

$$f(y_i) = \frac{1}{\mu_i} e^{-y_i/\mu_i}$$

The following data of medical insurance claims is available

Age	30	35	40	45	50
Claim amount(Rs '000)	50	80	150	250	360

The insurer believes that a linear function of age affects the claim amount i.e. $\eta_i = \alpha + \beta x_i$

wherein

x_i is the age of policyholder i

η_i is the claim amount of policyholder i

α and β are constants

- iv) Using the canonical link function, identify which one of the following equations gives the correct maximum likelihood estimates for α and β based on the above data.

A:

$$\frac{30\hat{\alpha}}{\hat{\alpha}+30\hat{\beta}} + \frac{35\hat{\alpha}}{\hat{\alpha}+35\hat{\beta}} + \frac{40\hat{\alpha}}{\hat{\alpha}+40\hat{\beta}} + \frac{45\hat{\alpha}}{\hat{\alpha}+45\hat{\beta}} + \frac{50\hat{\alpha}}{\hat{\alpha}+50\hat{\beta}} - 890 = 0$$

$$\frac{1}{\hat{\alpha}+30\hat{\beta}} + \frac{1}{\hat{\alpha}+35\hat{\beta}} + \frac{1}{\hat{\alpha}+40\hat{\beta}} + \frac{1}{\hat{\alpha}+45\hat{\beta}} + \frac{1}{\hat{\alpha}+50\hat{\beta}} - 39550 = 0$$

B:

$$\frac{1}{\hat{\alpha}+30\hat{\beta}} + \frac{1}{\hat{\alpha}+35\hat{\beta}} + \frac{1}{\hat{\alpha}+40\hat{\beta}} + \frac{1}{\hat{\alpha}+45\hat{\beta}} + \frac{1}{\hat{\alpha}+50\hat{\beta}} - 890 = 0$$

$$\frac{30\hat{\beta}}{\hat{\alpha}+30\hat{\beta}} + \frac{35\hat{\beta}}{\hat{\alpha}+35\hat{\beta}} + \frac{40\hat{\beta}}{\hat{\alpha}+40\hat{\beta}} + \frac{45\hat{\beta}}{\hat{\alpha}+45\hat{\beta}} + \frac{50\hat{\beta}}{\hat{\alpha}+50\hat{\beta}} - 39550 = 0$$

C:

$$\frac{1}{\hat{\alpha}+30\hat{\beta}} + \frac{1}{\hat{\alpha}+35\hat{\beta}} + \frac{1}{\hat{\alpha}+40\hat{\beta}} + \frac{1}{\hat{\alpha}+45\hat{\beta}} + \frac{1}{\hat{\alpha}+50\hat{\beta}} - 890 = 0$$

$$\frac{30}{\hat{\alpha}+30\hat{\beta}} + \frac{35}{\hat{\alpha}+35\hat{\beta}} + \frac{40}{\hat{\alpha}+40\hat{\beta}} + \frac{45}{\hat{\alpha}+45\hat{\beta}} + \frac{50}{\hat{\alpha}+50\hat{\beta}} - 39550 = 0$$

D:

$$\frac{30\hat{\alpha}}{\hat{\alpha}+30\hat{\beta}} + \frac{35\hat{\alpha}}{\hat{\alpha}+35\hat{\beta}} + \frac{40\hat{\alpha}}{\hat{\alpha}+40\hat{\beta}} + \frac{45\hat{\alpha}}{\hat{\alpha}+45\hat{\beta}} + \frac{50\hat{\alpha}}{\hat{\alpha}+50\hat{\beta}} - 890 = 0$$

$$\frac{30\hat{\beta}}{\hat{\alpha}+30\hat{\beta}} + \frac{35\hat{\beta}}{\hat{\alpha}+35\hat{\beta}} + \frac{40\hat{\beta}}{\hat{\alpha}+40\hat{\beta}} + \frac{45\hat{\beta}}{\hat{\alpha}+45\hat{\beta}} + \frac{50\hat{\beta}}{\hat{\alpha}+50\hat{\beta}} - 39550 = 0$$

E:

None of the above

(4)
[10]
